

The effect of interfacial elasticity on vesicle shape and cross-stream migration

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ABSTRACT We consider an extension of the model of Chan and Leal (*J. Fluid Mech.*, vol. 92, 1979, pp. 131–170) (1) for the migration of a fluid droplet in a parallel-plate channel that allows us to analytically solve the corresponding problem for a deformable vesicle with a simple Helfrich bending resistance. The close mathematical connection between the two systems allows for the direct solution of what would normally be complicated partial differential equations. The leading-order surface shape, particle velocity, and conditions leading to cross-stream migration are determined analytically for Poiseuille and Couette flows, and some mistakes in the formulas of the original paper are pointed out. We conclude with remarks on the implications of our theoretical results for the control of deformable vesicles in microfluidic systems.

SIGNIFICANCE Cells are frequently transported through confined fluid flows, and their ability to migrate across streamlines can strongly influence their biological function. Transverse migration plays an important role in biological processes including the margination of blood cells toward vessel walls, interactions between circulating cells and the endothelium, and the transport of cells through microvascular and porous environments. Here, we examine the dynamics of elastic vesicles flowing through walled channels and investigate how elastic interfacial properties and fluid stresses govern vesicle shape and transverse migration. Our results provide a simple framework for understanding the coupled dynamics of deformable membranes and confined flows, with implications for cellular transport and instrumentation in biological and engineered microfluidic systems.

INTRODUCTION

In many biological systems, cells and other particles experience confined fluid flows, where hydrodynamic stresses can drive migration across streamlines. Such transverse migration can strongly influence cellular transport and function, particularly near walls and obstacles (2–6). In the microcirculation, for instance, the lateral redistribution of blood cells and nanoparticles affects their margination toward vessel walls and their interactions with the vascular endothelium (7–10). These flow-induced dynamics can also be exploited for sorting, focusing, and separation in industrial and laboratory applications (11–13). In all of these settings, particle deformability can substantially impact the degree of migration (14–16). From an engineering perspective, this dependence on deformability suggests the related “inverse problem” of mechanical characterization via fluid flow (17, 18): transverse migration provides an observable that reflects the effects of cell deformability, which can be a marker for cellular health or pathology (19–21). Developing tractable theoretical models for the behavior of elastic particles in confined geometries may therefore help establish quantitative connections between flow measurements and the mechanical properties of cells.

The mechanisms underlying transverse migration depend on the mechanical properties of the transported object. For rigid particles, lateral migration in the absence of external forces is fundamentally an inertial effect (22–24) as Stokes reversibility forbids such motions. Deformable particles, by contrast, can migrate even in the zero-Reynolds-number limit because hydrodynamic stresses deform their interfaces, producing asymmetric shapes and associated lateral forces (25). This phenomenon has been studied for fluid droplets (1, 26, 27), elastic particles (28, 29), and viscoelastic particles (30). For cells and vesicles, the interface has a more complex mechanical response, but at a minimum, it possesses both a surface tension and a resistance to bending (31). Many studies of vesicles in fluid flows have focused on free flows or interactions with a single wall in a half-space (2, 3, 32, 33), whereas those in confined geometries tend to focus on the regime where the vesicle size is comparable to the channel width and confinement induces substantial stress and deformation (34–36). Relatively less attention has been given to the opposite limit of a small vesicle near two confining walls (37), where an analytical treatment can provide insight into the interplay between vesicle mechanics and hydrodynamic lift.

We are thus interested in analytically studying the deformation and lift of a small vesicle near two bounding walls. A natural starting point is the similar calculation of Chan and Leal (1) for the deformation and migration of a fluid droplet flowing through a parallel-plate channel. In this model, surface tension at the fluid interface provides the restoring force that controls the droplet shape. A deformable vesicle introduces an additional resistance to bending that is naturally described by the Helfrich model (31), in which the surface energy contains both tension and curvature-elastic contributions. Despite this added complexity, the two problems are mathematically closely related: both describe fluid interfaces whose shapes are determined by a balance between interfacial stresses and hydrodynamic forces, while the governing equation for a Helfrich membrane can be viewed as a natural higher-order generalization of the corresponding surface-tension problem (38–40), so that under appropriate regimes, their solutions are qualitatively similar. This connection provides an opportunity to extend established results for simple fluid droplets to elastic interfaces while retaining much of the analytical structure of the original problem.

Here, we amend the theoretical model proposed by Chan and Leal in (1) to include a Helfrich resistance to out-of-plane bending. We exploit the mathematical connection between Helfrich membranes and fluid droplets to derive the steady-state deformation and migration of a small elastic vesicle in a parallel-plate channel without expending significant additional effort. As in (1), we consider both Poiseuille and simple shear flows and determine the conditions for transverse migration, examining how interfacial tension, bending resistance, and hydrodynamic stresses influence the resulting lift. The formulation allows us to obtain analytical expressions for the surface shape and particle velocity while retaining the relative simplicity of the classical droplet problem. Potential applications of our results are briefly discussed.

METHODS

Fluid dynamical model

We consider a vesicle within an infinite channel consisting of two parallel plates subjected to a background quadratic flow, as shown in Fig. 1. In standard Cartesian coordinates (x, y, z) with corresponding unit basis vectors $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ the horizontal coordinate (parallel to the walls) is x , while the vertical coordinate (normal to the walls) is z . We will work primarily in spherical polar coordinates $\{r, \theta, \phi\}$, defined by the relation $(x, y, z) = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$, and corresponding basis vectors $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$. For simplicity, we work in the vesicle-center-of-mass frame. The flow is assumed to have zero Reynolds number Re so that the velocity and pressure fields are well described by the Stokes equations (41). In equations, if $\mathbf{u}$ is the velocity field outside of the vesicle and p is the corresponding pressure field,

$$-\nabla p + \mu \Delta \mathbf{u} = 0 \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Likewise, inside the vesicle,

$$-\nabla p' + \mu \Delta \mathbf{u}' = 0 \quad (3)$$

$$\nabla \cdot \mathbf{u}' = 0, \quad (4)$$

where $\mathbf{u}'$ and p' are the velocity and pressure fields inside the vesicle, respectively. It is assumed that the fluid cannot penetrate the vesicle interface and that the vesicle is neutrally buoyant within the channel fluid. For simplicity, we will take the fluid inside and outside the vesicle to be Newtonian so that the Cauchy stress tensors are

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu\mathbf{E} \quad (5)$$

$$\boldsymbol{\sigma}' = -p'\mathbf{I} + 2\mu'\mathbf{E}', \quad (6)$$

where $\mathbf{E} = (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)/2$ and similarly for $\mathbf{E}'$. We impose a quadratic background flow so that the far-field boundary condition can be written as

$$\mathbf{u} \rightarrow (\alpha + \beta z + \gamma z^2)\mathbf{e}_x - \mathbf{U} \text{ as } r \rightarrow \infty \quad (7)$$

for appropriately chosen constants α , β , and γ . Here, $\mathbf{U}$ is the translation velocity of the vesicle relative to the background flow, which is to be determined. The no-slip boundary condition on the walls is

$$\mathbf{u} = V_w \mathbf{e}_x - \mathbf{U} \text{ on } \mathcal{W} \quad (8)$$

where V_w is the speed at which the walls move.

There are also boundary conditions on the vesicle interface $\mathcal{S}$. Consistent with (1), we assume the following boundary conditions:

- Continuity of velocity:

$$\mathbf{u}|_{\mathcal{S}} = \mathbf{u}'|_{\mathcal{S}} \quad (9)$$

- Vanishing of normal velocity:

$$\mathbf{u} \cdot \mathbf{n}|_{\mathcal{S}} = \mathbf{u}' \cdot \mathbf{n}|_{\mathcal{S}} = 0 \quad (10)$$

- Balance of stress:

$$\boldsymbol{\sigma} \cdot \mathbf{n} - \boldsymbol{\sigma}' \cdot \mathbf{n}|_{\mathcal{S}} = -F\mathbf{n} \quad (11)$$

where F is the normal stress provided by the vesicle interface. To explore the effect of bending resistance, we assume that the vesicle deformation energy is given by the Helfrich bending energy (31) E_H plus the fluid-interfacial energy E_F ,

$$E_H = \frac{\kappa}{2} \oint_{\mathcal{S}} dA (2H)^2 + \bar{\kappa} \oint_{\mathcal{S}} dA K, \quad E_F = \eta \oint_{\mathcal{S}} dA, \quad (12)$$

where η is the surface tension, H is the mean curvature, and K is the Gaussian curvature. Then,

$$F = 2\kappa(\Delta_L B H - 2HK + 2H^3) - 2\eta H, \quad (13)$$

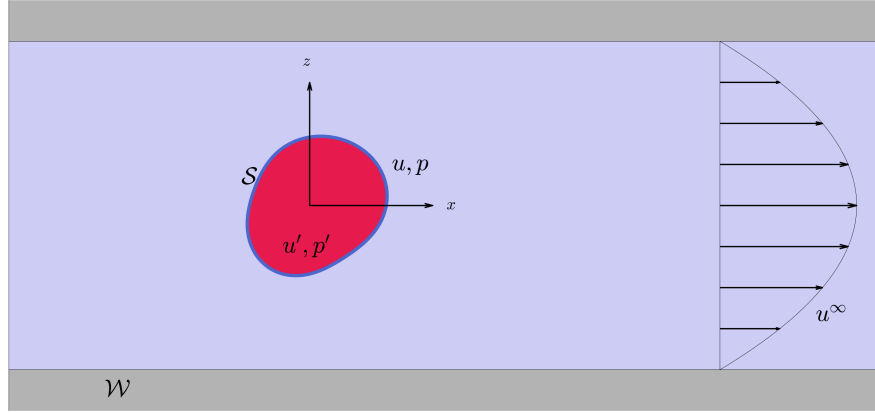


Figure 1: Schematic of the geometry considered in this study. A deformable vesicle (red) whose surface S is an elastic interface (dark blue) described by the Helfrich bending energy is placed in a parallel-plate channel. The outside fluid (light blue) satisfies no-slip conditions at the walls W . The flow field and pressure outside the vesicle are $\mathbf{u}$ and p , respectively, while inside the vesicle they are $\mathbf{u}'$ and p' , respectively. In this diagram, the background flow field $\mathbf{u}^\infty$ is parabolic.

where Δ_{LB} is the Laplace–Beltrami operator for the membrane surface, assumed to be a sphere initially. This expression is obtained by taking the first variation of $E = E_H + E_F$ (32). The moduli κ and $\bar{\kappa}$ describe the resistance of the surface to deformations that induce curvature. Note that F is independent of $\bar{\kappa}$; this is because if S remains closed, then the integral of the Gaussian curvature is a topological constant and therefore has vanishing variation. Thus, in addition to the usual surface tension term present in fluid droplets, we also have forces proportional to κ arising from bending the membrane. We naturally recover the droplet case by letting $\kappa \rightarrow 0$.

In our model, η is a prescribed constant, analogous to surface tension. This means that tangential Marangoni stresses are absent in our model. A common approximation is to assume that the Helfrich interface has a large areal compressibility modulus so that its surface area is fixed on the relevant time scales. In this formulation, η is not prescribed but rather is determined by enforcing the constraint of fixed area. We instead use the “ensemble” of fixed tension and variable area for two reasons. First, particles that move through microchannels can be moving quite fast and thus incur large viscous stresses (11, 17). It is unlikely that the assumption of fixed surface area holds in these situations. In addition, a particle that begins as a sphere and has both fixed surface area and fixed volume is “geometrically rigid:” there are no other shapes available to it because any deformation necessarily violates one of these two constraints, and the solution is trivial.

Equations 1–11 constitute a complicated boundary value problem with no simple analytical solution. In such cases, asymptotic analysis can be a useful tool for obtaining approximate (and hopefully accurate) solutions in certain parameter regimes. We will closely follow the theory of Chan and Leal (1) as a road map to obtain a perturbative solution to our problem and explore the effects of κ on vesicle dynamics.

Dimensionless groups

The primary determinant of membrane deformation will be the balance of forces in the normal direction. Forces in directions tangent to the surface may give rise to a tangential flow field, but this will not alter the shape. Thus, we examine the normal stress balance condition

$$\mathbf{n} \cdot (\boldsymbol{\sigma} - \boldsymbol{\sigma}') \cdot \mathbf{n} = -2\kappa(\Delta_{LB}H - 2HK + 2H^3) + 2\eta H \quad (14)$$

and nondimensionalize it. Suppose the membrane has an equilibrium radius of R . Let μ be the viscosity scale, R be the length scale, and the maximum velocity of the background Poiseuille flow U_{max} be the velocity scale. The pressure scale is $\mu U_{max}/R$ for the pressure outside and $\mu' U_{max}/R$ for the pressure inside, resulting in

$$\begin{aligned} \mathbf{n} \cdot \left[-\frac{\mu U_{max}}{R} \left(p' - \frac{\mu'}{\mu} p \right) + 2\frac{\mu U_{max}}{R} \left(E - \frac{\mu'}{\mu} E' \right) \right] \cdot \mathbf{n} \\ = -\frac{2\kappa}{R^3} (\Delta_{LB}H - 2HK + 2H^3) + \frac{2\eta}{R} H \end{aligned} \quad (15)$$

Rearranging, we have

$$\begin{aligned} \mathbf{n} \cdot [-(p' - \lambda p) + 2(E - \lambda E')] \cdot \mathbf{n} \\ = -\frac{1}{Be} 2(\Delta_{LB}H - 2HK + 2H^3) + \frac{1}{Ca} (2H), \end{aligned} \quad (16)$$

and our dimensionless parameters are

$$\lambda = \frac{\mu'}{\mu} \quad (\text{viscosity ratio}), \quad (17)$$

$$Be = \frac{\mu U_{max} R^2}{\kappa} \quad (\text{bending capillary number}), \quad (18)$$

$$Ca = \frac{\mu U_{max}}{\eta} \quad (\text{capillary number}). \quad (19)$$

We will work in the large elasticity/tension limit, $Be, Ca \ll 1$, which ensures that the membrane remains mostly spherical. Furthermore, we will assume $Be \sim Ca$; that is, both elasticity and tension play a role.

We will also find it useful to nondimensionalize with respect to the channel height h . Define

$$\zeta = \frac{R}{h} \quad (\text{radius-to-channel-width ratio}) \quad (20)$$

as well as the rescaled z -coordinate s such that $s = 0$ at the bottom wall and $s = 1$ at the top wall. If the coefficients α, β , and γ are expressed in terms of these parameters, we find that $\alpha = V_w s$, $\beta = V_w \zeta$, $\gamma = 0$ for Couette flow and $\alpha = 4U_{max}s(1-s)$, $\beta = 4U_{max}(1-2s)\zeta$, $\gamma = -4U_{max}\zeta^2$ for Poiseuille flow. Our results will be expressed as asymptotic series in both ζ and Ca . We describe how to formulate these series below.

Asymptotic expansion

To start, we let the membrane shape be perturbed from its base spherical geometry, with the order parameter of the perturbation amplitude being $\delta = Ca$. We then assume that the surface can be described in dimensionless units by

$$r(\theta, \phi) = 1 + \delta f^{(1)}(\theta, \phi) + \delta^2 f^{(2)}(\theta, \phi) + \dots \quad (21)$$

We similarly expand the velocity, pressure, and stress fields in δ :

$$\mathbf{u} = \mathbf{u}^{(0)} + \delta \mathbf{u}^{(1)} + \delta^2 \mathbf{u}^{(2)} + \dots \quad (22)$$

$$p = p^{(0)} + \delta p^{(1)} + \delta^2 p^{(2)} + \dots \quad (23)$$

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{(0)} + \delta \boldsymbol{\sigma}^{(1)} + \delta^2 \boldsymbol{\sigma}^{(2)} + \dots \quad (24)$$

For the inner quantities, we have a slightly different expansion:

$$\mathbf{u}' = \mathbf{u}'^{(0)} + \delta \mathbf{u}'^{(1)} + \delta^2 \mathbf{u}'^{(2)} + \dots \quad (25)$$

$$p' = \frac{1}{\delta} p'^{(-1)} + p'^{(0)} + \delta p'^{(1)} + \delta^2 p'^{(2)} + \dots \quad (26)$$

$$\boldsymbol{\sigma}' = \frac{1}{\delta} \boldsymbol{\sigma}'^{(-1)} + \boldsymbol{\sigma}'^{(0)} + \delta \boldsymbol{\sigma}'^{(1)} + \delta^2 \boldsymbol{\sigma}'^{(2)} + \dots \quad (27)$$

As we will see, the $O(\delta^{-1})$ terms are required to balance large stresses due to tension and bending in the quiescent case.

Keeping with the usual strategy for perturbations that deform geometry, we will pull all expanded quantities back to $r = 1$. For example, the surface normal in terms of quantities defined on $r = 1$ is

$$\mathbf{n} = \mathbf{e}_r - \delta \nabla_s f^{(1)} - \delta^2 \left(\frac{1}{2} |\nabla_s f^{(1)}|^2 \mathbf{e}_r + \nabla_s f^{(2)} - f^{(1)} \nabla_s f^{(1)} \right) + \dots, \quad (28)$$

where $\nabla_s f = \mathbf{e}_\theta \partial_\theta f + \mathbf{e}_\phi (\sin \theta)^{-1} \partial_\phi f$ is the surface gradient on the base unit sphere geometry. The mean and Gaussian

curvatures of the perturbed shape are calculated from the trace and the determinant of the Hessian matrix,

$$\begin{aligned} 2H|_S &= \nabla \cdot \mathbf{n} \\ &= 2 - \delta (\Delta_s + 2) f^{(1)} \\ &\quad + \delta^2 [-(\Delta_s + 2) f^{(1)} + 2(f^{(1)})^2 + 2f^{(1)} \Delta_s f^{(1)}] + \dots \end{aligned} \quad (29)$$

$$\begin{aligned} K|_S &= \frac{1}{2} [(\text{tr } \nabla \mathbf{n})^2 - \text{tr } (\nabla \mathbf{n})^2] \\ &= 1 - \delta (\Delta_s + 2) f^{(1)} \\ &\quad + \delta^2 \left[3(f^{(1)})^2 - \cot^2 \theta \csc^2 \theta (f_\phi^{(1)})^2 \right. \\ &\quad \left. + 2 \cot \theta \csc^2 \theta f_\phi^{(1)} f_{\theta\phi}^{(1)} - \csc^2 \theta f_{\theta\phi}^{(1)} + \csc^2 \theta f_{\phi\phi}^{(1)} f_{\theta\theta}^{(1)} \right. \\ &\quad \left. + \cot \theta f_{\theta\phi} f_{\theta\theta}^{(1)} + 3f^{(1)} \Delta_s f^{(1)} - (\Delta_s + 2) f^{(2)} \right] + \dots \end{aligned} \quad (30)$$

where Δ_s is the Laplacian on the unit sphere. Note that the $O(\delta)$ terms of $2H$ and K are both equal to $-(\Delta_s + 2)f^{(1)}$, which is due to the spherical base geometry. We define the differential operator in this term to be $\mathcal{L}_\infty = -(\Delta_s + 2)$. Interestingly, our expressions for $\mathbf{n}$ and $2H$ differ from those of (1) at $O(\delta^2)$, possibly due to different definitions of gradient. However, the discrepancy does not affect the results of (1), as the only term where they would appear, $F^{(1)}$, fortuitously does not enter the calculation at the orders considered. In principle, we must also expand the Laplace–Beltrami operator, as it depends on the geometry, but its perturbations are also part of terms that will ultimately vanish, so we will not consider these expressions here.

Letting $Be = B\delta$ so that $B = Be/Ca = O(1)$ and following the same asymptotic notation as above, we find that

$$F^{(-1)} = 2, \quad F^{(0)} = \mathcal{L}_B f^{(1)} := \frac{1}{B} (\Delta_s - B)(\Delta_s + 2) f^{(1)}. \quad (31)$$

In the $B \rightarrow \infty$, which describes an elasticity-free droplet, $\mathcal{L}_B \rightarrow \mathcal{L}_\infty = -(\Delta_s + 2)$, and we recover the linear Jacobi operator from (1) in $F^{(0)}$.

The expanded quantities are substituted into Eqns. 1 to 13, and a hierarchy of problems is generated, one for each power of δ starting with δ^{-1} . In the following, we will discuss the solutions to the problems at orders δ^{-1} , δ^0 , and δ^1 , allowing us to determine the shape and velocity of the vesicle to leading order.

The $O(1/\delta)$ problem

We begin by matching terms of $O(\delta^{-1})$. Since the flow field has no such terms, this situation represents the quiescent case $\mathbf{u} = \mathbf{u}' = \mathbf{0}$, which trivially satisfies the continuity and kinematic boundary conditions at the sphere surface and the no-slip condition at the walls. The only nontrivial boundary condition is

$$-\boldsymbol{\sigma}'^{(-1)} \cdot \mathbf{n}^{(0)} = (2H^{(0)}) \mathbf{n}^{(0)}. \quad (32)$$

In this case, $\sigma'^{(-1)} = -\lambda p'^{(-1)} \mathbf{I}$, which naturally yields the Young–Laplace law:

$$p'^{(-1)} = \frac{2}{\lambda}, \quad (33)$$

where the 2 is the leading order contribution to $2H$ for a sphere. In writing this equation, we have set the pressure to zero outside the membrane, $p^{(-1)} = 0$.

The $O(1)$ problem

In principle, the calculation proceeds by matching terms proportional to $O(1)$. However, at $O(1)$ the flow field is nonzero, and so the enforcement of boundary conditions both at the walls and the vesicle surface becomes significantly harder. Chan and Leal (1) addressed this challenge by using the method of reflections (41), a relative of the well-known method of images for constructing solutions to Stokes flows with symmetric boundary conditions. The basic idea behind the method of reflections is to first solve the free-space problem without walls. Additional terms are then added that approximately account for the boundary conditions at $\mathcal{W}$ and $\mathcal{S}$. In particular, we assume the outer flow field is well approximated by $\mathbf{u}^{(0)} = \mathbf{u}_f^{(0)} + \mathbf{u}_w^{(0)} + \mathbf{u}_s^{(0)} + \dots$, where $\mathbf{u}_f^{(0)}$ is the free-space solution (i.e., ignoring the no-slip boundary condition on the walls), $\mathbf{u}_w^{(0)}$ is added to enforce the no-slip condition on the walls (i.e., $\mathbf{u}_w^{(0)} + \mathbf{u}_f^{(0)}|_{\mathcal{W}} = 0$), and $\mathbf{u}_s^{(0)}$ is added to enforce the kinematic boundary condition on the sphere (i.e., $\mathbf{u}_s^{(0)} + \mathbf{u}_w^{(0)}|_{\mathcal{S}} = 0$). The inner flow field is constructed in a similar manner. In general, an infinite series of terms is necessary to impose the wall boundary condition exactly, yielding a power series in ζ . If $\zeta \ll 1$, it is valid to truncate this series to obtain an approximate solution; our truncated solution will be accurate to $O(\zeta^4)$. The calculation of $\mathbf{u}_w^{(0)}$ from $\mathbf{u}_f^{(0)}$ and $\mathbf{u}_s^{(0)}$ from $\mathbf{u}_w^{(0)}$ in the infinite channel geometry can be carried out using Faxén’s method of Fourier transforms (41, 42).

While the procedure described above is tried and true, it is also quite tedious. Before discussing it further, we pause to make an important observation: the modification of the normal stress boundary condition to include Helfrich stresses, Eqn. 11, does not affect the $O(1)$ solutions for the flow and pressure fields derived by Chan and Leal. This is because only the tangential continuity condition, the kinematic boundary condition, and the *tangential* stress balance are used to solve for the full flow field in this geometry. All of these boundary conditions are identical to the ones in our case, Eqns. 9–11—only the *normal* stress balance is different. Thus, we are able to directly use the solution for the flow field derived in (1) without modification. For this reason, and for clarity of presentation, we will omit the details of the procedure used to find the flow field and refer the reader interested in the full expressions to the papers by Chan and Leal and also Ho and Leal (42). The flow fields are given in Eqns. 4.5–4.9 of (1) in terms of 26 constants $A_1, \dots, H_1, A_3, \dots, H_3$,

$A'_1, \dots, H'_1, A'_3, \dots, H'_3$ whose values are tabulated in Eqns. 4.11–4.12; note that there is a typo in Eqn. 4.9: $-3\tilde{E}_1$ should be positive, not negative. The expressions for the definite integrals I_1, I_2, I_4, I_5, I_7 that appear in the wall corrections resulting from Faxén’s method can be found in (42). These integrals depend on the position of the particle within the channel and are thus generally functions of s . They are also all proportional to ζ^3 . If the net force on the vesicle is to be zero (a requirement for neutrally buoyant objects in steady-state), then the x -velocity of the particle at $O(1)$, $U^{(0)}$, is constrained, as given by Eqn. 4.14 of (1), which we reproduce here:

$$U^{(0)} = \alpha + \frac{\lambda\gamma}{2 + 3\lambda} + I_1 + \frac{1}{2}I_4. \quad (34)$$

At this order of accuracy, the vesicle’s bending stiffness thus has no impact on its speed. As I_1 and I_4 depend on s , Eqn. 34 relates the particle speed to its position in the channel. With the flow field determined, the solution to the differential equation Eqn. 11 provides the deformation $f^{(1)}$.

The $O(\delta)$ problem

The results up to this point do not predict any cross-streamline migration, so we proceed to examine the $O(\delta)$ problem. In (1), Chan and Leal applied the reciprocal theorem for Stokes flows (41, 43), allowing them to bypass the explicit calculation of the flow and pressure fields at this order and obtain the transverse migration velocity directly. We once again find that many of their results are applicable to our system and show how to modify their derivation to include the effect of finite B here.

We begin by finding the $O(\delta)$ boundary conditions at $r = 1$, which we will need to use the reciprocal theorem result of (1). By expanding Eqns. 9–11, we obtain

$$\mathbf{u}^{(1)} + f^{(1)} \frac{\partial \mathbf{u}^{(0)}}{\partial r} \Big|_{r=1} = \mathbf{u}'^{(1)} + f^{(1)} \frac{\partial \mathbf{u}'^{(0)}}{\partial r} \Big|_{r=1} \quad (35)$$

$$\begin{aligned} \mathbf{u}^{(1)} \cdot \mathbf{n}^{(0)} + \mathbf{u}^{(0)} \cdot \mathbf{n}^{(1)} + f^{(1)} \frac{\partial (\mathbf{u}^{(0)} \cdot \mathbf{n}^{(0)})}{\partial r} \Big|_{r=1} \\ = \mathbf{u}'^{(1)} \cdot \mathbf{n}^{(0)} + \mathbf{u}'^{(0)} \cdot \mathbf{n}^{(1)} + f^{(1)} \frac{\partial (\mathbf{u}'^{(0)} \cdot \mathbf{n}^{(0)})}{\partial r} \Big|_{r=1} = 0 \end{aligned} \quad (36)$$

$$\begin{aligned} \sigma^{(1)} \cdot \mathbf{n}^{(0)} + \sigma^{(0)} \cdot \mathbf{n}^{(1)} + f^{(1)} \frac{\partial (\sigma^{(0)} \cdot \mathbf{n}^{(0)})}{\partial r} \\ - \lambda \left[\sigma'^{(1)} \cdot \mathbf{n}^{(0)} + \sigma'^{(0)} \cdot \mathbf{n}^{(1)} + f^{(1)} \frac{\partial (\sigma'^{(0)} \cdot \mathbf{n}^{(0)})}{\partial r} \right. \\ \left. + f^{(1)} \frac{\partial (\sigma'^{(-1)} \cdot \mathbf{n}^{(1)})}{\partial r} \right] \Big|_{r=1} = -F^{(0)} \mathbf{n}^{(1)} - F^{(1)} \mathbf{n}^{(0)} \Big|_{r=1} \end{aligned} \quad (37)$$

Note that in Eqn. 37, the $\sigma'^{(-1)} \cdot \mathbf{n}^{(2)}$ term and $F^{(-1)} \mathbf{n}^{(2)}$ term have been eliminated using Eqn. 33. However, there is still a term that involves the derivative of $\sigma^{(-1)} \cdot \mathbf{n}^{(1)}$, which is not necessarily nonzero. This term does not seem to have been considered in (1) but does not appear to have a large effect on the results, so we will similarly drop it for consistency with (1).

The above boundary conditions at $r = 1$ are rearranged and substituted into Eqn. 6.4 of (1), which we reproduce below in our notation:

$$U^{(1)} = -\frac{1 + \lambda}{2\pi(2 + 3\lambda)(1 + J_1 + J_4)} \oint_{r=1} dA \left[\mathbf{v} \cdot (\sigma^{(1)} - \lambda \sigma'^{(1)}) - \mathbf{u}^{(1)} \cdot (\mathbf{t} - \lambda \mathbf{t}') - \lambda (\mathbf{u}^{(1)} - \mathbf{u}'^{(1)}) \cdot \mathbf{t}' \right] \cdot \mathbf{e}_r, \quad (38)$$

where $\mathbf{v}$, $\mathbf{t}$, and $\mathbf{t}'$ are the velocity, outer stress, and inner stress fields of the complementary solution, respectively. Expressions for the integrals J_1 and J_4 (which are both $O(\zeta)$) are given in (42). This equation is highly general, and the derivation makes no assumptions about the interfacial stress so that it is valid for our system. By substituting Eqns. 35-37 into this equation, we obtain an expression for the migration velocity that does not require explicit knowledge of $\mathbf{u}^{(1)}$ or $\mathbf{u}'^{(1)}$. In (1), the flow field due to a drop moving in the z -direction was chosen as the complementary problem. The expressions for these quantities are given in Eqns. 4.17–4.25 of (1), which we use. The reader is cautioned that Eqn. 4.24b of (1) appears to contain a typo: we believe that the $-5r^2 \mathbf{e}_3$ term should read $-5zr^2 \mathbf{e}_3$ to satisfy incompressibility. Once again, because the only difference between our system and that of (1) is the normal stress balance, our results are expected to be qualitatively similar.

RESULTS AND DISCUSSION

Steady-state vesicle shapes

We now turn to the calculation of the membrane shape, which is determined by the normal stress balance Eqn. 16 and is expected to differ from the droplet case. At $O(1)$, the perturbed shape of the boundary is determined by solving an equation involving the linear differential operator $\mathcal{L}_B$. From the factored form Eqn. 31, we observe an important mathematical connection between the droplet and vesicle cases: the operator $\mathcal{L}_B = -(1/B)(\Delta_s - B)\mathcal{L}_\infty$ is also diagonalized by ℓ -th order spherical harmonics $Y_\ell^{\pm 1}$. Thus, the solution of the shape is facilitated by decomposing the stress jump across the interface into spherical harmonics. We remark that the operator $-(\Delta_s - B)/B$ is positive definite for $B > 0$ and thus does not introduce resonances.

Since the left-hand side of Eqn. 11 is independent of B , rather than explicitly calculating the difference in normal stress at $r = 1$, which would require detailed calculation of the flow field and pressure, we simply use Chan and Leal's

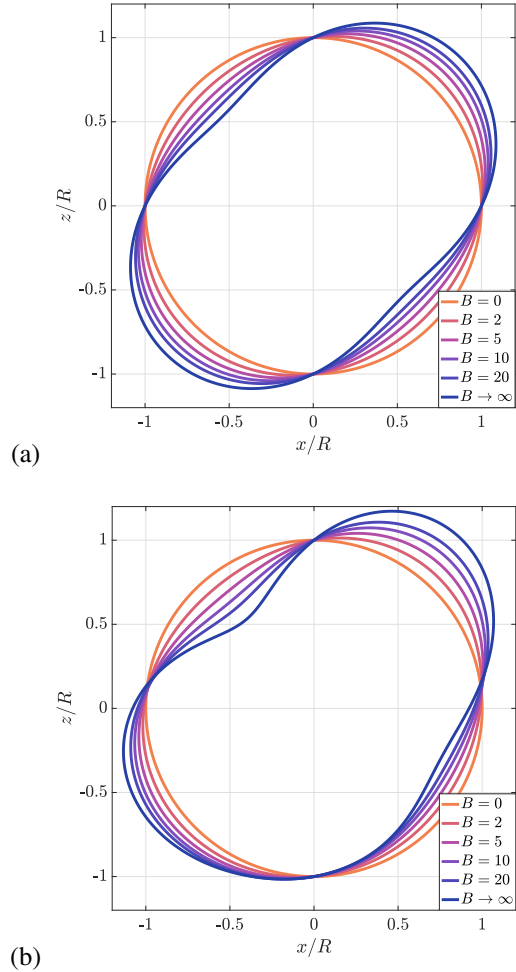


Figure 2: Dependence of vesicle shape on bending stiffness $B = Be/Ca$. (a) Shear flow ($\beta = 1$, $\alpha = \gamma = 0$). (b) Poiseuille flow ($\alpha = 0$, $\beta = 1$, $\gamma = 1.5$). In both panels, $Ca = \delta = 0.2$, $\lambda = 1$, and $\zeta = 0.2$.

solution for the droplet deformation,

$$f_\infty^{(1)} = \frac{16 + 19\lambda}{8(1 + \lambda)} \left[\beta - \zeta \left(\frac{3}{2} I_2 + I_5 \right) \right] \frac{xz}{r^2} - \frac{\gamma(10 + 11\lambda)}{40(1 + \lambda)} \left(\frac{x_1}{r} - \frac{5xz^2}{r^3} \right), \quad (39)$$

and determine the normal traction jump at the interface by solving $\mathcal{L}_B f^{(1)} = \mathbf{e}_r \cdot (\sigma - \lambda \sigma') \cdot \mathbf{e}_r|_{r=1} = \mathcal{L}_\infty f_\infty^{(1)}$. (Note that Eqn. 4.16b in (1) contains $(2/3)I_2$ instead of $(3/2)I_2$; this is likely a misprint given the form of the coefficients in Eqn. 4.12.) Since $f_\infty^{(1)}$ consists of only an $\ell = 2$ mode and an $\ell = 3$ mode, the solution is straightforward to write down. Using $\Delta_s Y_\ell^{\pm 1} = -\ell(\ell + 1)Y_\ell^{\pm 1}$, we see that

$$\mathcal{L}_B Y_\ell^{\pm 1} = \frac{1}{B} [-\ell(\ell + 1) - B] [-\ell(\ell + 1) + 2] Y_\ell^{\pm 1}. \quad (40)$$

Thus, $f^{(1)}$ is also a linear combination of $\ell = 2$ and $\ell = 3$ modes:

$$f^{(1)} = \frac{B}{B+6} \frac{16+19\lambda}{8(1+\lambda)} \left[\beta - \zeta \left(\frac{3}{2} I_2 + I_5 \right) \right] \frac{xz}{r^2} - \frac{B}{B+12} \frac{\gamma(10+11\lambda)}{40(1+\lambda)} \left(\frac{x_1}{r} - \frac{5xz^2}{r^3} \right), \quad (41)$$

and the effect of B is to modulate the harmonic amplitudes with prefactors that lie between 0 (rigid limit, $B \rightarrow 0$) and 1 (droplet limit, $B \rightarrow \infty$). The machinery of (1, 42) has allowed us to calculate the shape of a vesicle in a channel with minimal modification. Again, this is to be expected because the Helfrich model has a mathematical structure very similar to that of the droplet problem.

As expected, decreasing the bending stiffness of the interface increases the deformation of the vesicle. As $Be \rightarrow \infty$, the droplet shape from (1) is recovered. In the opposite limit $Be = 0$, the droplet interface completely resists out-of-plane deformation and the shape remains spherical. (However, as Be increases, it is also increasingly less likely that an interface will wrap itself into a vesicle in the first place (44).) As shown in Fig. 2, when Be is set to a finite value, the shape interpolates between the shapes these in two limits. Shapes with large B can have regions of mean curvature significantly greater than 1 in magnitude, including regions where convexity of the shape is lost, while for shapes with small B , this curvature is suppressed.

Similar to droplets, reducing the viscosity of the inner fluid in a vesicle increases the deformation. However, altering the viscosity does not have a significant effect on the shape as the prefactors $(16+19\lambda)/[8(1+\lambda)]$ and $(10+11\lambda)/[40(1+\lambda)]$ in $f^{(1)}$ do not vary strongly with λ . Proximity to either wall (s closer to 0 or 1) also increases the deformation through the parameter β . This phenomenon is consistent with experimental images of vesicles near channel walls, such as those appearing in (45).

Migration in Poiseuille flow

We evaluate Eqn. 38 to find the migration velocity as a function of s , ζ , λ , and B . The full expressions up to $O(\zeta^4)$ are complicated, so we compute these numerically. However, to $O(\beta\gamma) = O(\zeta^3)$, we find that for Poiseuille flow

$$U^{(1)} = - \frac{B\beta\gamma [36P_1(\lambda) + BP_2(\lambda)]}{210(6+B)(12+B)(1+\lambda)^2(4+\lambda)(2+3\lambda)^2}, \quad (42)$$

where

$$P_1(\lambda) = 560 + 33\lambda - 2592\lambda^2 - 2697\lambda^3 - 587\lambda^4 + 33\lambda^5 \quad (43)$$

$$P_2(\lambda) = 2320 + 1843\lambda - 6292\lambda^2 - 7327\lambda^3 - 1242\lambda^4 + 198\lambda^5. \quad (44)$$

We have verified that the $B \rightarrow \infty$ limit matches the result in (1). Conversely, as the particle becomes more rigid ($B \rightarrow 0$), the

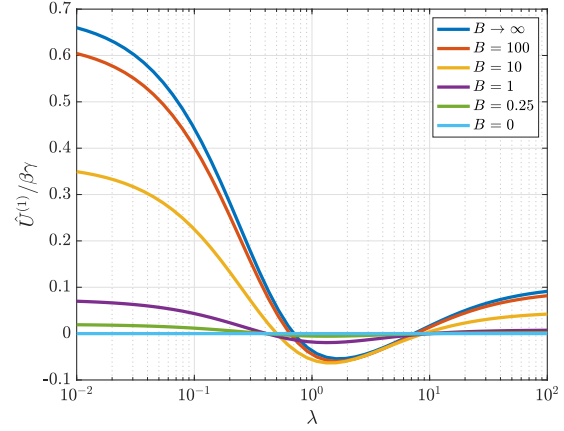


Figure 3: Leading order transverse migration velocity of a vesicle as a function of B and λ in Poiseuille flow. Decreasing B causes the speed to decrease uniformly in λ .

migration velocity goes to zero, as a rigid particle remains spherical and thus experiences no lift in Stokes flow. Figure 3 illustrates $U^{(1)}/(\beta\gamma)$ as a function of λ for different B values using a linear-log scale. Regardless of B , $U^{(1)}/(\beta\gamma)$ is positive for sufficiently high or sufficiently low λ , whereas a range of intermediate λ always leads to a negative $U^{(1)}/(\beta\gamma)$ at this order. In particular, the case of equal inner and outer viscosities is predicted to always lead to a negative migration velocity (normalized by $\beta\gamma$).

The expression Eqn. 42, which is a simple affine function of s , does not explicitly contain any of the integrals from Faxén's method, meaning that the effects of the walls only materialize at higher order. In Fig. 4, we plot the full $O(\zeta^4)$ expression for $U^{(1)}$ as a function of channel position $s \in [0, 1]$ to analyze these wall effects. (Regions close to either wall at $s = 0$ or 1 are omitted as the method of reflections is inaccurate there and predicts a diverging velocity.) We see that the dependence on B is nonmonotonic in certain portions of the graph; a similar phenomenon was noted in (45) for vesicles of varying reduced volume.

This plot can also be viewed as the stability diagram of the autonomous ordinary differential equation (ODE) $ds/dt = U^{(1)}$, which approximately governs the height of the particle in the channel. Simple stability analysis can thus be used to predict the steady-state positions s^* of vesicles in the channel. For $\zeta = 0.1, \lambda = 1$ (Fig. 4a), there are three steady-state solutions to this ODE. In the exact middle of the channel, the migration velocity vanishes by symmetry. This steady-state position is unstable, as perturbing s leads to a velocity that pushes the particle further away from the center. In addition, there are two stable nodes where $U^{(1)} = 0$ located at approximately $s = 0.35$ and 0.65 . As B increases, these nodes move further away from the wall, although this effect is small for the set of parameters in Fig. 4a (varying B by three orders of magnitude only moves s^* by approximately

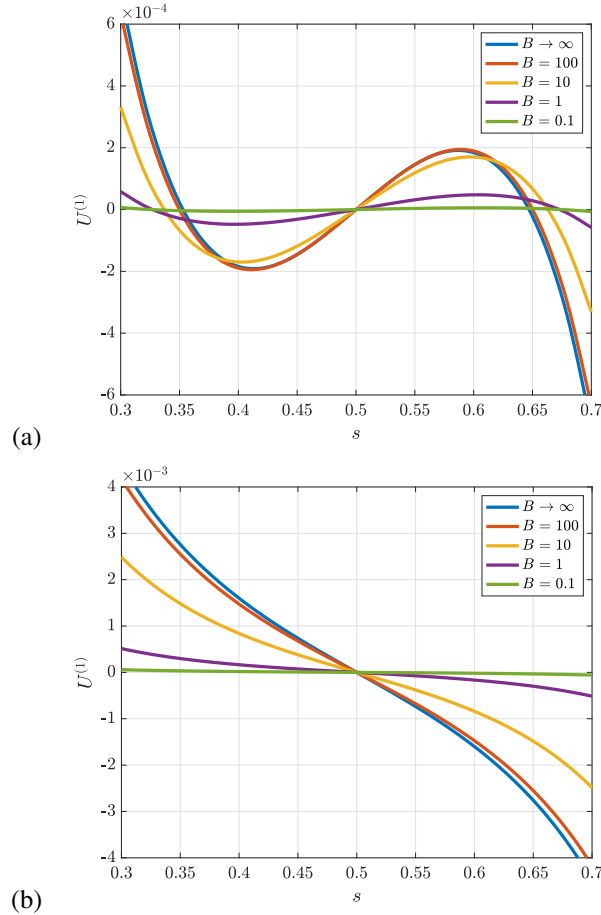


Figure 4: Migration velocity as a function of channel position s in a Poiseuille flow. (a) $V_{max} = 1$, $\zeta = 0.1$, $\lambda = 1$. There are three steady-state positions where $U^{(1)} = 0$. The node at $s = 0.5$ is unstable to small perturbations while the other two are stable. (b) $V_{max} = 1$, $\zeta = 0.1$, $\lambda = 0.1$. There is only one stable steady-state position at the channel center.

$0.05h$). We find that decreasing ζ moves these stable locations closer to the walls, perhaps allowing for a broader range of observable positions. Since the flow speed is greater towards the channel center, softer vesicles are thus predicted to not only drift further away from the walls but also move faster in the x -direction. Given an initial position, a prediction of vesicle trajectories is possible by integrating the vector $\mathbf{U}(s) = U^{(0)}(s)\mathbf{e}_x + \delta U^{(1)}(s)\mathbf{e}_z$ in time, assuming that the trajectory remains within the region of validity of this expression.

For any B , if the viscosity ratio λ is increased or decreased sufficiently from $\lambda = 1$, a pitchfork bifurcation occurs and only one steady-state node at $s^* = 0.5$ remains (Fig. 4b). This node is stable, meaning that in cases of high viscosity contrast, vesicles are predicted to naturally flow towards the channel center. The monotonic dependence of $U^{(1)}$ on B is restored in these cases.

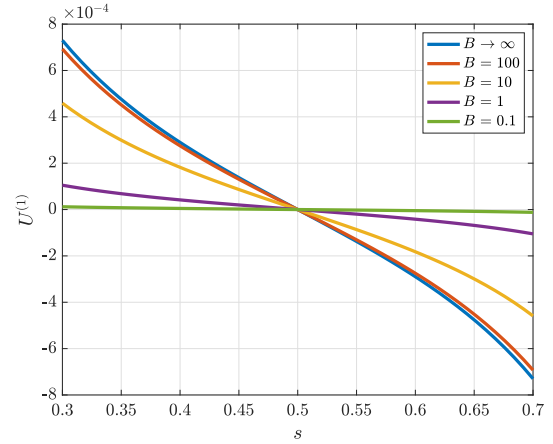


Figure 5: Leading order migration velocity as a function of channel position s in a Couette flow. $V_w = 1$, $\zeta = 0.1$, $\lambda = 1$. The channel center ($s = 0.5$) is a stable steady-state position.

Migration in Couette flow

For Couette flow, $\gamma = 0$, and the leading order contribution to the migration velocity is of $O(\zeta^4)$:

$$U^{(1)} = \frac{3\zeta B J_2 \beta^2 (16 + 19\lambda)(9 + 17\lambda + 9\lambda^2)}{140(B + 6)(1 + \lambda)^2(2 + 3\lambda)} \quad (45)$$

which originates from the leading order contribution of $f^{(1)}$. This formula predicts that the velocity decreases with increasing λ . For B , we see again that $B = 0$ yields $U^{(1)} = 0$ and that increasing B increases $U^{(1)}$ monotonically. In the limit of $B \rightarrow \infty$, this result differs very slightly from that of (1); we were unable to find the source of the discrepancy, but the difference is basically negligible. However, our prefactor notably agrees with that in (46, 47) (with the erroneous factor of 33 replaced by 3, as explained in (1)). The term J_2 , much like J_1 and J_4 , is an integral that depends on s and is defined in Eqn. 4.22 of (1); it is an odd function of s with respect to $s = 0.5$. For $s < 1/2$, the dominant term of J_2 is proportional to $1/s^2$, so that our Eqn. 45 for the migration velocity possesses the same scaling form as that in (3, 48), although the vesicle interfacial elasticity was not explicitly considered in these references. Figure 5 shows how migration velocity depends on s in a Couette flow up to $O(\zeta^4)$. Similar to Fig. 4b, $U^{(1)} > 0$ for $s < 1/2$ and is odd about $s = 0.5$. Hence, $s^* = 0.5$ is the sole steady-state position, is stable, and will attract vesicles toward it regardless of B or λ .

CONCLUSION

With simple adjustments, the model of Chan and Leal (1) has allowed us to analyze the hydrodynamics of a vesicle in a flow between two plates. The Helfrich model turns out to be very compatible with this calculation, and the elasticity, quantified by the parameter Be , interpolates between the two cases of the droplet (maximum deformation and lift) and

the rigid particle (zero deformation and lift). Depending on parameters, there can be multiple preferred lateral positions for vesicles in Poiseuille flow, while there is generally only one at the channel center in Couette flow. Since the height of the vesicle depends on Be , and since the speed in the channel depends on height, vesicles could theoretically be sorted by elasticity, or the elasticity of a vesicle could potentially be used to control its trajectory, which could be of use in applications involving targeted drug delivery or microswimmers. From the perspective of an inverse problem, these observables could form the foundation of a method to infer Helfrich elasticity.

The minimal model considered here neglects many physical effects for the sake of analytical tractability and is restricted to parameter regimes with small ζ , Be , and Ca . In particular, we do not account for inertia, cellular activity, thermal fluctuations, interfacial viscosity, or Marangoni stresses, each of which may influence particle dynamics under appropriate conditions. For example, inertial forces can drive the migration of rigid particles (22) and may provide additional lift at sufficiently high particle velocities. Similarly, cellular activity and external fields have been shown to enhance margination in blood flow (49). Incorporating these effects, along with the complex internal structure of living particles, will be necessary to develop more comprehensive models and improve predictions for biological applications.

AUTHOR CONTRIBUTIONS

L.L.J. designed the research, carried out the research, and wrote the article.

DECLARATION OF INTERESTS

The author declares no competing interests.

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